Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1 *Fill in the following blanks with the correct choice of the words from this list:*

Increasing, decreasing, positive, negative, concave up, concave down

(a) *If you know $f''(x) > 0$, then you know $f'(x)$ is* ____________________ *and $f(x)$ is* ____________________.

(b) *If you know $g'(x) < 0$ and decreasing, then you know $g(x)$ is* ____________________ *and* ____________________.

(c) *If you know $h(x)$ is positive, increasing, and concave down, then you know $h'(x)$ is* ____________________ *and* ____________________ *and that $h''(x)$ is* ____________________.

[4]

Problem 2

February 21, 2026

Problem 2 *Sketch a graph of a function that is continuous on* $(-\infty, \infty)$ *that has the following properties.*

(a) *Function* f *does not have a local maximum or minimum.* f *contains a point where* $f'(x) = 0$

(b) $g'(x) < 0$ *on* $(-\infty, -1)$; $g'(x) > 0$ *on* $(-1, 2)$; $g'(x) < 0$ *on* $(2, \infty)$.

Problem 3

February 21, 2026

Problem 3 *Give an example or sketch of a function that is continuous on* $(-\infty, \infty)$ *and satisfies given properties. If such a function does not exist, explain why.*

(a) *A function* f *is concave up and negative everywhere.*

(b) *A function* f *is decreasing and concave up everywhere.*

(c) *A function* s *has exactly 3 local extrema and four inflection points.*

(d) *A function* f *has exactly 2 zeros and one local extrema.*

Problem 4

February 21, 2026

Problem 4

(a) *You are given that* $f''(x) > 0$ *for all* x*. Which of the following must be true about* $f(x)$ *on the region* $0 \leq x \leq 2$*?*

(i) *There is a critical point between* 0 *and* 2*.*

(ii) *There is a local maximum, but not enough information is given to determine where.*

(iii) f *need not have a local maximum.*

(b) *You are told that* $f''(x) > 0$ *for all* x*. Which of the following must be true about the graph of* $y = f(x)$*?*

(i) *The graph is a straight line.*

(ii) *The graph crosses the* x*-axis at most once.*

(iii) *The graph is concave down.*

(iv) *The graph crosses the* y*-axis more than once.*

(v) *The graph is concave up.*

Problem 5

February 21, 2026

Problem 5 *Suppose a function f satisfies the following conditions:*

(a) $f(0) = 0$ *and* $f'(-4) = f'(2) = f'(10) = 0$

(b) $\lim_{x \to 6} f(x) = -\infty$, *and* $\lim_{x \to +\infty} f(x) = 6$

(c) $f'(x) < 0$ *on* $(-\infty, -4)$, $(2, 6)$, *and* $(10, +\infty)$

(d) $f'(x) > 0$ *on* $(-4, 2)$, *and* $(6, 10)$

(e) $f''(x) > 0$ *on* $(-\infty, 0)$, *and* $(14, +\infty)$

(f) $f''(x) < 0$ *on* $(0, 6)$, *and* $(6, 14)$

(a) *List the* ***interval(s)*** *where the function f is* ***both increasing*** *and* ***concave UP***.

(b) *List the* ***interval(s)*** *where the function f is* ***both increasing*** *and* ***concave DOWN***.

(c) *List the* ***interval(s)*** *where the function f is* ***both decreasing*** *and* ***concave UP***.

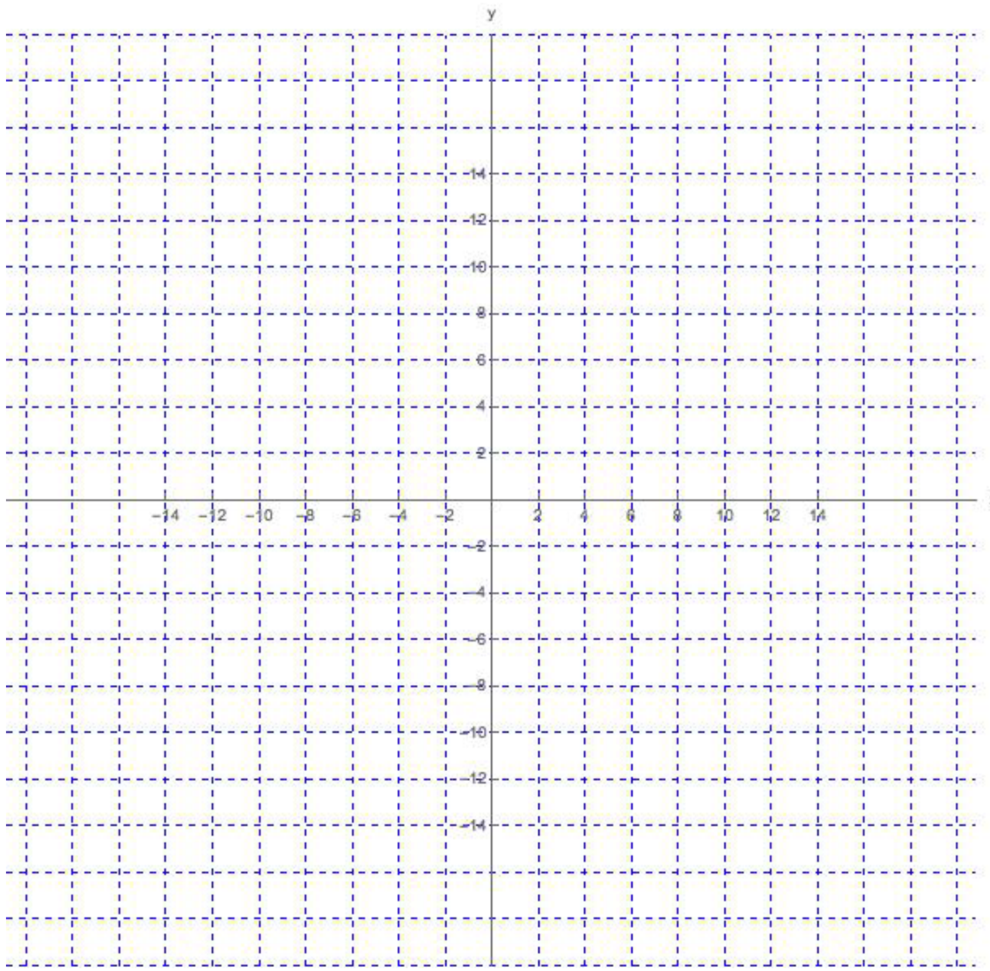
(d) *List the* ***interval(s)*** *where the function f is* ***both decreasing*** *and* ***concave DOWN***.

(e) *List the* ***x-coordinates*** *at which f has a* ***local minimum***. *Write ”none” if appropriate.*

(f) *List the* ***x-coordinates*** *at which f has a* ***local maximum***. *Write ”none” if appropriate.*

(g) *List the* ***x-coordinates*** *of all* ***inflection points*** *of f. Write ”none” if appropriate.*

(h) *Sketch the graph of f.*


y
x
−14 −12 −10 −8 −6 −4 −2 2 4 6 8 10 12 14
14 12 10 8 6 4 2 −2 −4 −6 −8 −10 −12 −14

Problem 6

February 21, 2026

Problem 6 *Sketch the graph of a function f satisfying all of the conditions:*

(a) f *is continuous and* odd, $f(0) = 0$,

(b) $\lim_{x \to \infty} f(x) = -5$,

(c) $f'(x) > 0$ *on* $(6, \infty)$,

(d) $f'(x) < 0$ *on* $(0, 6)$,

(e) $f''(x) > 0$ *on* $(0, 12)$, *and*

(f) $f''(x) < 0$ *on* $(12, \infty)$.